

# Tokenization in Python (Detailed Study Notes)

## 1 Introduction

Tokenization Natural Language Processing (NLP) ka ek basic aur bohat important step hai. Iska matlab hai text ko chhote meaningful parts (tokens) me divide karna. Yeh tokens words, sentences ya characters ho sakte hain.

Roman Urdu me: Tokenization ka matlab hai kisi bhi text ko tukron me todna taake computer usko easily samajh sake.

## 2 Types of Tokenization

- Word Tokenization
- Sentence Tokenization
- Character Tokenization
- Regex-based Tokenization

## 3 Word Tokenization

Word tokenization sentence ko individual words me divide karta hai.

### Examples

#### Example 1

```
from nltk.tokenize import word_tokenize
word_tokenize("Python is easy")
```

Explanation: Sentence ko 3 words me divide kiya.

#### Example 2

```
word_tokenize("Hello world!")
```

Explanation: Punctuation bhi alag token ban gaya.

#### Example 3

```
word_tokenize("AI is future")
```

Explanation: Har word alag token hai.

#### Example 4

```
word_tokenize("Data Science amazing hai")
```

Explanation: Mixed language bhi tokenize hoti hai.

#### Example 5

```
word_tokenize("Python 3.10 best hai")
```

Explanation: Numbers aur punctuation bhi tokens me aate hain.

#### Example 6

```
word_tokenize("Machine learning powerful hai")
```

Explanation: Har word separate ho jata hai.

#### Example 7

```
word_tokenize("Deep learning amazing hai")
```

Explanation: Har word ek token hai.

#### Example 8

```
word_tokenize("Coding fun hai")
```

Explanation: Simple words tokenize ho gaye.

#### Example 9

```
word_tokenize("Practice daily")
```

Explanation: Do tokens ban gaye.

#### Example 10

```
word_tokenize("Learn Python fast")
```

Explanation: Har word alag ho gaya.

## 4 Sentence Tokenization

Sentence tokenization text ko sentences me divide karta hai.

### Examples

#### Example 1

```
from nltk.tokenize import sent_tokenize  
sent_tokenize("Python easy hai. Bohat useful hai.")
```

Explanation: Do sentences me split kiya.

#### Example 2

```
sent_tokenize("Hello. How are you?")
```

Explanation: Full stop aur question mark pe split.

#### Example 3

```
sent_tokenize("AI future hai. Seekho!")
```

Explanation: Exclamation pe bhi split hota hai.

#### Example 4

```
sent_tokenize("Data science popular hai. Jobs bhi hain.")
```

Explanation: Multiple sentences detect hue.

#### Example 5

```
sent_tokenize("Coding seekho. Practice karo. Grow karo.")
```

Explanation: Teen sentences ban gaye.

#### Example 6

```
sent_tokenize("ML powerful hai. Time lagta hai.")
```

#### Example 7

```
sent_tokenize("NLP useful hai. Text analyze karta hai.")
```

#### Example 8

```
sent_tokenize("Python simple hai. Beginner friendly hai.")
```

#### Example 9

```
sent_tokenize("Projects banao. Skills improve karo.")
```

#### Example 10

```
sent_tokenize("Learn daily. Grow fast.")
```

## 5 Character Tokenization

Character tokenization text ko individual characters me todta hai.

### Examples

#### Example 1

```
list("Python")
```

Explanation: Har letter separate ho gaya.

#### Example 2

```
list("Hello")
```

#### Example 3

```
list("AI")
```

#### Example 4

```
list("Data")
```

#### Example 5

```
list("Code")
```

#### Example 6

```
list("ML")
```

#### Example 7

```
list("NLP")
```

#### Example 8

```
list("Learn")
```

#### Example 9

```
list("Grow")
```

#### Example 10

```
list("Fast")
```

## 6 Regex-based Tokenization

Regex ka use karke bhi tokenization ki ja sakti hai.

### Examples

#### Example 1

```
import re
re.findall(r"\w+", "Hello, kaise ho?")
```

Explanation: Punctuation remove karke sirf words mil gaye.

#### Example 2

```
re.findall(r"\w+", "Python123")
```

Explanation: Words aur numbers ek sath token bane.

#### Example 3

```
re.findall(r"\w+", "AI & ML")
```

Explanation: Special characters remove ho gaye.

#### Example 4

```
re.findall(r"\w+", "Data-science")
```

Explanation: Hyphen remove ho gaya.

#### Example 5

```
re.findall(r"\w+", "Email: abc@gmail.com")
```

Explanation: Email parts me break ho gaya.

#### Example 6

```
re.findall(r"\w+", "Score: 90%")
```

#### Example 7

```
re.findall(r"\w+", "Time 12:30")
```

#### Example 8

```
re.findall(r"\w+", "Price $100")
```

#### Example 9

```
re.findall(r"\w+", "Hello!!!")
```

#### Example 10

```
re.findall(r"\w+", "Good_morning")
```

## 7 Conclusion

Tokenization NLP ka basic step hai jo text ko understandable format me convert karta hai. Yeh machine learning aur text analysis ke liye foundation provide karta hai.